ERTS Lab
Department of Computer Science and Engineering
Indian Institute of Technology Bombay, Powai
Mumbai - 400 076.

Certificate of Participation

This is to certify that **Aditya Shivanand Mhamane**, a student of **PDPM Indian Institute of Information Technology, Design and Manufacturing, Madhya Pradesh** has participated in the **e-Yantra Robotics Competition (eYRC 2025-26)**.

He/She is a member of the team having the following team members,

1. Podila Gouri Nandana Krishna
2. Suchendra Charith Gondi
3. Aditya Shivanand Mhamane
4. A A Jai Abishek

This team has participated in **KrishiBalancer** theme.

Prof. Shivaram Kalyanakrishnan
Principal Investigator, e-Yantra
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[P.T.O]

eYantra
Engineering a better tomorrow

Theme: *KrishiBalancer*

Theme Description:

This is to certify that the team has successfully designed, developed, and implemented a PID/LQR-based control system that enables a self-balancing robot to maintain stability in the upright position.

The robot is to initiate operation from a designated starting point and is capable of traversing in the arena with path while performing remote pick-and-place operations of fertilizers using a mobile Bluetooth application. Based on the configuration provided the robot accurately identifies, picks up, and delivers the specified fertilizer to its designated location. To evaluate the robustness and effectiveness of the implemented control strategy, various types of obstacles are introduced along the path, challenging the robot's balance, navigation, and overall control performance.

Participants are required to independently design and fabricate the balancing robot, as well as implement appropriate control algorithms such as PID or LQR to achieve stable balancing behavior.In addition, participants must design and integrate a suitable manipulator mechanism to enable reliable pick-and-place functionality, thereby demonstrating a comprehensive integration of mechanical design, control theory, and embedded system implementation.

The theme is executed in simulation and hardware. The simulation is carried out in Coppeliasim.

Theme Learning: *PID/ LQR Controller, Mathematical modeling, Python, Build-a-bot/Mechanical Design, Embedded systems*

Certificate Level: The team has been awarded **Level 3** certificate in the e-Yantra Robotics Competition (eYRC).

> ***Level 3 indicates:*** *The team showed a fair understanding of the problem statement by implementing a partial solution of the problem statement.*

Certificate Level Matrix

Certificate Level	Certificate Description
Level 1	**Certificate of Merit**
Level 2	Certificate of Completion
Level 3	**Certificate of Participation**
Level 4	Letter of Participation

* Level 1 > Level 2 > Level 3 > Level 4 (Level 1 is highest achievement)